

Exactly 21 experimental resources enter a numerical analysis

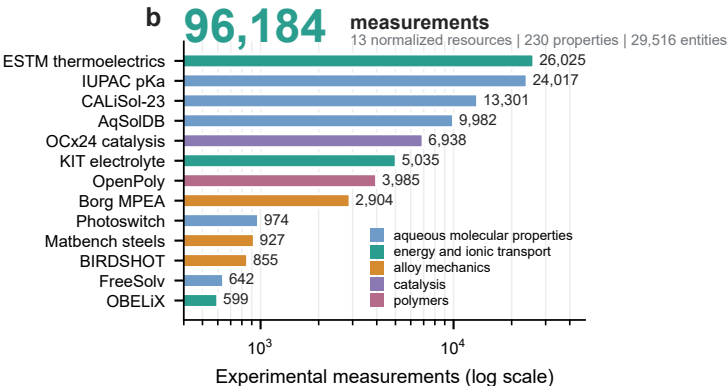
a

Twenty form the directed borrowing network; one supplies the independent artifact gate

RESOURCE	SCIENTIFIC FAMILY	DONOR	RECIPIENT	GATE	LAYER
DONOR + RECIPIENT (13)					
AqSolDB	aqueous molecular	●	●	●	N
FreeSolv	aqueous molecular	●	●	●	N
CALiSol-23	electrolyte transport	●	●	●	N
ESTM thermoelectrics	thermoelectric transport	●	●	●	N
KIT electrolyte conductivity	electrolyte transport	●	●	●	N
Borg MPEA	alloy mechanics	●	●	●	N
OBELiX	ionic transport	●	●	●	N
OCx24	electrocatalysis	●	●	●	N
OpenPoly	polymers	●	●	●	N
Photoswitch	photochemistry	●	●	●	N
Caltech ionic conductivity	ionic transport	●	●	●	E
Multi-stage battery aging	battery aging	●	●	●	E
SpecGen derivative OER	electrocatalysis	●	●	●	E
DONOR ONLY (3)					
IUPAC digitized pKa	aqueous molecular	●	●	●	N
Caltech Acid-OER	electrocatalysis	●	●	●	E
Caltech metal-oxide ORR	electrocatalysis	●	●	●	E
RECIPIENT ONLY (4)					
BIRDSHOT	alloy mechanics	●	●	●	N
Matbench steels	alloy mechanics	●	●	●	N
Starrydata2	thermoelectric transport	●	●	●	E
TRI four-plate OER	electrocatalysis	●	●	●	E
ARTIFACT GATE ONLY (1)					
NIST ISODB	adsorption thermodynamics	●	●	●	A

N = normalized | E = frozen external | A = streamed analysis-only

The source-pinned measurement core



c The empirical denominator is explicit

21 analysed resources
13 normalized | 7 external | 1 analysis-only

20 transfer-active resources
16 donors | 17 recipients | 13 both

97 systematic gate-benchmark edges
20 target tasks

13 programme clusters
positive, null, harmful and unresolved retained

Favorable and unfavorable outcomes stay in the same evidence portfolio

d

	Law / coefficient transport	Few-shot prediction	OOD screening	Sequential / temporal
Meyer-Neldel thermoelectrics	×	●	●	●
ISODB adsorption	▲	●	●	●
Borg to BIRDSHOT	×	●	●	●
Borg to Matbench	●	×	●	●
FreeSolv to AqSolDB	●	×	●	●
KIT -20 to -30 C	●	●	●	●
ESTM to OBELiX	●	●	●	×
CALiSol -30 to -40 C	●	▲	●	●
OBELiX and ESTM to Caltech	●	●	●	×
Ionic sources to Starrydata	●	▲	●	●
OER ORR and OCx to TRI	●	×	●	●
Battery Stage 1 to Stage 2	●	●	●	◆
SpecGen OER B and D	●	●	●	●
SpecGen OER A and C	●	▲	●	●

● gate passed ▲ unresolved / conditioned × null, harm or non-evaluable
● directional / internal signal ◆ method-development diagnostic ■ bold row: role-defining example